Mini Project Report

Department of Computer Science and Engineering

Abstract Submission

Abstract 1
Secure Military Messaging Application with End-to-End Encryption and Self-Destructing Messages
Based on: IIT Jammu Hackathon

Abstract 2
Digital Platform for Centralized Alumni Data Management and Engagement
Based on: Smart India Hackathon (SIH)
Problem Statement ID: 25017
Organization: Government of Punjab
Department: Department of Higher Education

Abstract 3
Mobile Application for Material Testing and Quality Check
Based on: Gujarat Government Hackathon
Problem Statement ID: PS000070
Organization: Gujarat Urja Vikas Nigam (GUVNL)

Submitted To

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Date of Submission: 05 December 2025

Secure Military Messaging App with Local-Only Storage and End-to-End Encryption

Mini Project Report

Team Members:
Abhiram P.
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Abstract

Building a simple secure messaging prototype that demonstrates how private communication can be safeguarded without depending on server-side storage is the main goal of this small project. Nowadays, the majority of messaging apps store user messages or logs on centralized servers, which creates a risk in the event that those servers are compromised or monitored. The objective of this project is to develop an intuitive end-to-end encrypted chat system in which all message data is stored entirely on the user's device and is promptly deleted from the server upon delivery. This method keeps previous conversations from being retrieved by outsiders and lessens the possibility of data leaks. The mini project primarily demonstrates the fundamental idea of encrypted communication and lays the groundwork for more sophisticated features that will be developed in subsequent phases, including metadata protection and self-destructive messages.

Digital platform for Centralized Alumini relationship and management System

Mini Project Report

Team Members:
Abhiram P.
Antony S Kannampuzha
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Abstract

This phase is all about laying the groundwork for the alumni management system. We willl dive into analyzing requirements, designing the system architecture, and creating a secure, centralized database for alumni. Key features will include user registration, authentication, and role-based access for alumni, students, faculty, and administrators, along with storing personal, academic, and career information. Weâ€™ll make sure to prioritize data accuracy, security, and scalability to guarantee that the system remains reliable for years to come.

Mobile Application For Material Testing And Quality Check

Mini Project Report

Team Members:
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Geo Jose

Abstract

This phase is all about getting a solid grasp on the quality assurance workflows at the field level and turning GUVNL's material testing standards into a digital format. We'll nail down the functional requirements by chatting with stakeholders, and then move on to designing the system architecture and database. We'll be developing key modules like secure user authentication, role-based access for engineers, inspectors, and supervisors, and real-time data entry for test results, measurements, and observations. Plus, we'll lay the groundwork for standardized reporting and centralized data storage to make sure everything is accurate, traceable, and compliant.